

NABU COMMAND CENTER

MASTER PRODUCTION ENGINEERING WHITE PAPER

Architecture • Ground-Up Construction • Native Z80 Client • Integrated Gateway • Golden Transport • Live Telemetry
• Audio • Verification • Deployment • Release Engineering

Version 1.0 Production Record and Reproducible Build Manual

Document Revision 1.1 - Codex Technical Audit Applied

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26 August 2026

Native software for the NABU Personal Computer

Figure 1. Current production architecture: contemporary data is reduced upstream and delivered as bounded NABU-native records.

Current release-candidate evidence	Value
BUILD_ID	NCC-SPLASH-260826-LV2
000001.bin / 000001.NABU	59,501 bytes each; byte-identical
SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
MAME	VERIFIED by Owner regression
Physical NABU	OWNER VERIFIED on exact LV2 binary; physical interconnect details not yet frozen into a deployment manifest
Final Git freeze	PENDING - current LV2 is owner-accepted but not committed/tagged
Current resource gate	BSS_END \$FC7A; initial SP \$FF00; 646-byte static-to-stack margin

Table 1. Current production baseline at this document revision.

Current-build evidence: Owner regression and Codex build reports, 26 Aug 2026. Historical status in earlier papers is superseded where this current evidence is stronger.

Document Control, Evidence Boundary, and Supersession

Field	Current value
Document title	NABU Command Center - Master Production Engineering White Paper
Document purpose	Complete v1.0 software production engineering record and ground-up software reconstruction manual; third-party physical interconnect reconstruction requires Appendix L.
Product version	NABU Command Center v1.0
Document revision	1.1 / 26 August 2026
Owner / Chief Architect	Derek Leger aka (Super_Derek)
Primary target	NABU Personal Computer
Development host	Windows 11-class host; verified paths documented in this paper

Field	Current value
Native toolchain	z88dk 2.4 baseline; native +nabu target
Emulator	MAME 0.250 / nabupc
Runtime data path	NABU Internet Adapter RetroNET Store
Gateway	Single integrated Command Center Gateway
Current release-candidate	NCC-SPLASH-260826-LV2 - OWNER-ACCEPTED, UNFROZEN WORKING-TREE BASELINE
Physical status	OWNER VERIFIED on current LV2 binary; physical interconnect manifest remains incomplete for third-party replication
Technical audit	Codex end-to-end source/repository review applied; 9 MUST corrections and 9 SHOULD updates dispositioned in this revision.

Table 2. Master document control.

This paper consolidates and updates the August 7 engineering white paper, the August 14 engineering/status paper, the August 25 v1.0/v2.0 roadmap, and the Music Stream engineering paper. Those documents remain historical engineering evidence; their older status tables do not override later build, MAME, or physical-NABU evidence. [S08] [S09] [S10] [S11]

Evidence terms are used literally: VERIFIED means directly supported by source/build/artifact, MAME, Adapter, Gateway, or physical-hardware evidence at the stated layer; DESIGN COMPLETE means specified but not necessarily executed; SOURCE CREATED does not imply successful target build; NOT TESTED means the relevant gate was not run; INFERRED is a reasoned conclusion short of direct proof; FAILED means an attempted condition was proven not to work. MAME and physical NABU remain separate evidence gates. [S01] [S02]

The current product baseline is stronger than the predecessor papers: dynamic ZIP/location, live Airspace/ADS-B, truthful Task Manager telemetry, production Music, the startup splash, the maximized-Weather stack-corruption fix, integrated MAME regression, and physical-NABU regression have all advanced since the August 25 roadmap snapshot. The current Help expansion was intentionally not implemented because the measured 646-byte static-to-stack margin makes a dense 6-8 page help manual unsafe without a separate memory decision. This revision also distinguishes the owner-accepted LV2 runtime baseline from repository freeze: LV2 is verified in MAME and on physical NABU, but remains an unfrozen working-tree state until an Owner-authorized commit/tag/manifest is created.

Authority model

Role	Engineering authority
Derek	Owner and Chief Architect; owns vision, architecture, scope, acceptance, freeze, release, and phase advancement.
ChatGPT	Engineering Manager; evidence synthesis, planning, source review, task specification, defect triage, and acceptance recommendations.
Codex	Local implementation/build engineer and peer technical authority for source-level decisions; edits authorized files, runs focused tests/builds, and reports evidence.

Table 3. Project authority and execution roles.

Source hierarchy

1. Current physical-NABU evidence recorded for the exact artifact.

2. Current MAME acceptance evidence for the exact artifact.
3. Current production source for implementation semantics.
4. Current generated map/listing/symbol/binary/hash evidence for artifact identity and memory addresses.
5. Current Gateway source, tests, configuration, and launcher evidence.
6. Current Git commits, tags, manifests, and phase-state evidence.
7. Installed z88dk/NABU-LIB source, headers, libraries, configuration, and examples.
8. Original NABU and component-manufacturer documentation.
9. Earlier Command Center white papers, roadmaps, and historical project records.
10. External sources only where a claim is version-sensitive or not resolved by project evidence.

Documentation-review authority hierarchy: current source controls implementation semantics unless contradicted by higher runtime evidence; map/binary evidence controls artifact identity and measured addresses. [S01] [S02] [S16]

Master Contents

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Executive Summary

NABU Command Center is a native Z80 information and telemetry console that treats the NABU Personal Computer as a deterministic command terminal rather than a modern web client. Contemporary Internet complexity - HTTPS, JSON, API authentication, provider retries, geographic calculations, normalization, catalog parsing, and MIDI conversion - remains on a modern Gateway. The NABU receives compact records through the Internet Adapter's RetroNET Store path, validates them, preserves last-known-good state, and renders a dense command-center interface using the machine's Z80A, TMS9918A, keyboard, and AY-3-8910. [S03] [S04] [S08] [S09]

The most important architectural result is the frozen Golden Transport foundation. The project reproduced and localized an earlier 63/64-byte receive failure to the retired application-owned communications path, then rebased application work on a proven NABU-LIB/RetroNET Store pattern. Qualification completed 5,000/5,000 bounded reads with zero missing/truncated reads and zero freezes in its MAME gate. That transport decision is closed unless

stronger contradictory evidence appears. [S09] [S12]

The current v1.0 release candidate contains the six Page-1 systems - Atomic Clock, Local Weather, Earthquake, Space Weather, Satellite/ISS, and Airspace/ADS-B - plus NWS alerts, rolling weather history, dynamic ZIP/location resolution, three-page NABU System / Task Manager diagnostics, global 60-second refresh, independent manual R refresh, background Music, fixed startup splash/music, Sound controls, Help/Diagnostics/Safe-Idle views, and the complete owner-approved graphical presentation. Current build NCC-SPLASH-260826-LV2 compiled and linked, produced byte-identical 59,501-byte .bin/.NABU artifacts, and was Owner-verified in MAME and on physical NABU hardware.

This document is intentionally both an engineering record and a software construction manual. A competent engineer should be able to start with the preserved environment, prove the Hello Derek baseline, reproduce the z88dk build pipeline, configure the Internet Adapter and MAME, reconstruct the Golden transport, implement the Gateway publication pattern, add module caches and renderers one at a time, and qualify Music and the splash without undocumented conversation history. The Owner-verified physical result is recorded, but third-party physical interconnect reconstruction additionally requires the hardware deployment manifest in Appendix L; missing electrical/serial details are never inferred.

Figure 2. Recommended ground-up construction order. Each milestone is frozen before the next subsystem is added.

PART I - Product History, Scope, and Engineering Rules

1. Origin and product objective

The project began as a December 2023 Weather/News concept and evolved into a WOPR-inspired but original operations console. The idea that survived every architectural revision is simple: the modern host acquires and reduces information; the NABU remains the responsive display-and-control endpoint. Earlier papers describe the progression from a speculative BASIC concept to a native +nabu application, verified Store transport, live time/weather, an owner-frozen GUI, and ultimately the integrated v1.0 system. [S08] [S09]

The command-center aesthetic is an information-architecture choice, not a reproduction of a film interface. The production system uses original vector/grid layouts, compact text, status colors, scopes, trends, and economical AY audio. No safety-of-life, aviation-navigation, maritime-navigation, emergency-command, weapons, or autonomous-control function is part of the product. [S08]

2. Version 1.0 product boundary

Category	Version 1.0 content
Primary Page-1 modules	Atomic Clock; Local Weather; Earthquake; Space Weather; Satellite / ISS; Airspace / ADS-B
Weather enhancements	NWS alerts; rolling trends; ZIP-based location profile
Global services	60-second automatic refresh; manual R refresh; Sound; Music; persistent header/status
Diagnostics	Three-page NABU System / Task Manager; Diagnostics view; basic Help; Safe Idle
Startup	Owner-approved v1.0 splash; fixed embedded marching tune; 10-second maximum; any-key bypass
Deployment	MAME 0.250 and physical NABU operation
Release engineering	Build IDs, hashes, local Git freezes, release manifest, reproducible documentation

Category	Version 1.0 content
Deferred to v2.0	Dashboard paging, six additional data modules, expanded full-page diagnostics/help unless separately authorized

Table 4. Version 1.0 release boundary. Earlier roadmap status is updated by current evidence. [S10]

3. Engineering rules that do not change

- BUILD UPON ONLY. Preserve accepted GUI, controls, transport, layout, behavior, and frozen milestones.
- Do not redesign, simplify, reinterpret, modernize, or replace owner-frozen presentation without explicit Owner approval.
- One integrated Gateway owns modern Internet/API work.
- One Golden NABU transport owner serves the application; no per-module network stacks.
- Use bounded reads, fixed arrays, deterministic state machines, compact integer data, and last-known-good caches.
- Do not invent APIs, ports, addresses, return values, wiring, COM ports, or physical behavior.
- Verify the visible BUILD_ID before diagnosing runtime behavior.
- Manual R and automatic refresh are independent; manual refresh must not reset/postpone the automatic timer.
- Treat MAME and physical NABU as separate evidence gates.
- Freeze successful milestones before adding the next subsystem.
- After two failed attempts on one theory, stop and gather new evidence.

Owner-frozen design rules and production doctrine: [S01] [S12].

PART II - NABU Hardware Platform and Verified Foundation

4. Platform facts that constrain the design

Subsystem	Verified fact	Engineering consequence
CPU	Z80A; 10.738635 MHz crystal divided by three -> 3.579545 MHz CPU clock	Integer arithmetic, compact state machines, bounded per-frame work.
CPU RAM	64 KB dynamic RAM available to CPU	Static/fixed buffers; avoid dynamic allocation and large stack objects.
Video	TMS9918A; 16 KB separate VRAM, accessed through the VDP	256x192 Mode-2 vector/text presentation; bounded redraw regions.
Audio / I-O	AY-3-8910, three channels; two 8-bit parallel ports used by NABU interrupt/status logic	Audio code must preserve register-7/I-O semantics; channel ownership is explicit.
Keyboard	Detached keyboard via NABU hardware/IOS; current native client uses nonblocking getk()	Input remains responsive during scheduler, audio, and network operations.
Network heritage	HCCA/Adapter-oriented, network-loaded applications	Gateway/Adapter architecture is historically aligned, but current RetroNET semantics must be version-verified.

Table 5. NABU platform facts and design consequences. [S03] [S04]

The NABU Technical Manual documents the Z80A, 64K RAM, TMS9918A with separate 16K video RAM, AY-3-8910, keyboard interface, HCCA interface, and vectored interrupts. The original application-programming technical specification documents the network-oriented software model, TMS9918 programming environment, keyboard/IOS concepts, and application interfaces. [S03] [S04]

Figure 3. Measured memory constraint for the current hardware-verified release candidate. This margin is a release-engineering gate, not free RAM telemetry.

5. Hello Derek and Hardware Test Suite baselines

Before Command Center work, Hello Derek established a practical keyboard-first NABU C baseline. It deliberately reaches the input loop before unverified PSG or hardware-sprite calls, then exercises getk(), console positioning/colors, bounded delays, direct get_psg()/set_psg(), a four-note AY chime, and TMS9918 sprite bring-up. The source preserves AY register-7 I/O-direction bits rather than calling a blanket psg_init(). [S05]

The Hardware Test Suite supplements that baseline with fixed screen regions, numbered menus, raw key reporting, text/color tests, bounded software animation, system-information display, reset instructions, and safe return behavior. It is a diagnostic pattern, not a replacement for Hello Derek. [S07]

Baseline	Artifact / role	Status
HELLO DEREK v4	Keyboard, console, AY, sprite and build pipeline	Frozen known-good baseline
Hello Derek 000001.bin/.NABU	7,948 bytes; byte-identical; SHA-256 400FE3D93DAD95E0ABA2FD109D184D5510A67EAC2DF8C2A2C73D96798869CFE6	Verified binary evidence
Hardware Test Suite	Structured diagnostics and bounded menu/test pattern	Supplemental verified source baseline

Table 6. Preserved practical baselines. [S02] [S05] [S07]

PART III - Ground-Up Host and Toolchain Construction

6. Verified Windows paths

Purpose	Verified path
Primary workspace	<NABU_WORKSPACE>
Custom programs	<NABU_WORKSPACE>
Master repository	<publication-repository>
Active production client	<publication-repository>\client\phase3a_live_local_weather_v0_1\NABU_COMMAND_CENTER_LIVE_LOCAL_WEATHER_v0_1
Integrated Gateway	<publication-repository>\gateway\phase3a_live_local_weather_v0_1\NABU_COMMAND_CENTER_LOCAL_WEATHER_GATEWAY_v0_1
z88dk	<Z88DK_ROOT>
MAME 0.250	<MAME_ROOT> NABU 9938\mess.exe
Internet Adapter	C:\Program Files (x86)\DJ Sures\NABU Internet Adapter\NABU-Internet-Adapter-84.exe
RetroNET Store	D:\NABU Internet Adapter\Store

Purpose	Verified path
Git	C:\Program Files\Git\cmd\git.exe

Table 7. Verified local development and runtime paths. [S06] [S12]

7. Reproducible +nabu build pattern

The production BUILD.BAT retains the verified Z: mapping, PATH and ZCCCFG pattern but the integrated LV2 build has one additional source input and evidence-generating flags. The current native command is authoritative for reconstruction; the older Hello Derek command remains the baseline pattern only. [S06] [S16]

```
@echo off
setlocal
set "Z88DK_REAL=<Z88DK_ROOT>"
if not exist Z:\bin\zcc.exe subst Z: "%Z88DK_REAL%"
set "PATH=Z:\bin;%PATH%"
set "ZCCCFG=Z:\lib\config"
cd /d "%~dp0"
del /q 000001.bin 2>nul
del /q 000001.NABU 2>nul
del /q *.map 2>nul
del /q *.lis 2>nul
del /q *.sym 2>nul
"Z:\bin\zcc.exe" +nabu main.c minimal_vdp_isr.asm -create-app -m -s --list -o000001.bin
if errorlevel 1 exit /b 1
if not exist 000001.NABU exit /b 1
```

For a fresh engineering environment, prove the frozen Hello Derek pipeline first, then build the production client with the exact integrated command above. Current BUILD.BAT still prints the stale console label `Building NCC-RF-260826-LV1`; that echo is not build identity. Until separately corrected, main.c BUILD_ID plus generated artifact inspection controls identity. The production flags `-m`, `-s`, and `--list` are required because the map/symbol/listing evidence is used for the memory and release records in this paper. [S16]

8. Build evidence required for every production candidate

Evidence	Required result
Compiler output	No fatal compilation error
Linker output	Successful link; inspect map for memory/resource changes
Artifacts	000001.bin and 000001.NABU both present
Identity	Embedded BUILD_ID visible in source and binary
Binary relationship	Byte-identical .bin/.NABU when expected by current target behavior
Integrity	SHA-256 recorded
Map/listing	Retained for symbol/resource diagnosis
Runtime	MAME acceptance after build; physical separately

Table 8. Minimum build evidence packet.

Figure 4. Selected artifact-size growth from baseline smoke test to the current integrated v1.0 release candidate. The final dip reflects splash-era memory recovery, not lost product scope.

PART IV - System Architecture and Golden Transport

9. One Gateway, one transport owner, one scheduler

The production architecture is layered rather than feature-specific. Every live module contributes one Gateway acquisition/normalization path, one bounded Store resource, one NABU validator/cache, and one summary/detail renderer. It does not create another HCCA implementation, independent socket, parallel scheduler, or second Gateway. [S09] [S12]

Figure 5. System architecture reused by all v1.0 modules.

10. Retired 63/64 transport failure

The earlier mixed interrupt/HCCA design produced a reproducible receive pathology: sufficiently long critical/ISR latency delayed HCCA receive service, a UART overrun replaced or lost a pending byte, a nominal 64-byte operation returned only 63 bytes, and a blocking reader could wait forever for the missing byte. This was diagnosed rather than hand-waved away. The strategic correction was to stop modifying the transport theory and rebase application work onto proven NABU-LIB/RetroNET behavior. [S09] [S12]

Figure 6. Retired 64-byte / 63-byte freeze chain and the production architectural decision.

11. Golden Transport foundation

Item	Frozen evidence
BUILD_ID	NCC-TF-260813-GOLDEN-01
SHA-256	FA075985BB98DCAE8A4428F12FABD07139A9C4F6627F3622D986A5626CDD4ABD
Commit	bd5cb00692b567c6095b503d4c94f4230fda233f
Tag	ncc-transport-foundation-v1-mame-verified
Qualification	5,000/5,000 bounded reads
Missing/truncated	0
Freezes	0
MAME	VERIFIED

Table 9. Golden Transport freeze evidence. [S09] [S12]

TCP 5816 is program loading only. Runtime module data uses the Adapter's RetroNET Store service, historically associated with TCP 5815 in the verified configuration. The two ports have different purposes and must not be conflated. [S08] [S12]

Typical current read pattern (exact declarations come from installed z88dk/NABU-LIB):

```
handle = rn_fileOpen(name_length, name, OPEN_FILE_FLAG_READONLY, 0xff);
received = protected_file_read(handle, buffer, 0, read_offset, length);
rn_fileHandleClose(handle);
accept only when received == required_length; then validate before cache commit.
```

Current client source demonstrates `rn_fileOpen / protected_file_read / rn_fileHandleClose` on time, weather, earthquake, satellite, airspace and music paths. [S13]

12. Protected read and interrupt discipline

The integrated client wraps `rn_fileHandleRead` in `protected_file_read()`: interrupts are disabled, the current NABU interrupt mask is captured, all sources are masked except HCCA receive during the bounded read, then the previous mask is restored. This is current production source evidence; it is not a generic rule for every z88dk NABU application. [S13]

Future developers should reuse the exact installed-source-compatible wrapper and must not extrapolate APIs from unrelated z88dk targets. If installed declarations, success values, or close semantics differ in a new environment, stop and inspect them before changing production transport. [S01]

PART V - Integrated Gateway and Bounded Record Publication

13. Gateway responsibilities

Gateway responsibility	NABU responsibility
HTTPS/API acquisition, authentication, provider-specific headers	Request bounded Store resources
JSON/XML/GeoJSON/MIDI parsing	Validate fixed record structure and checksum
Unit conversion, sanitization, sorting and top-N selection	Commit complete valid state only
Geospatial and orbital calculations	Render ready-to-display values
Rate limits, retry/circuit-breaker behavior	Remain responsive during provider failure
Atomic record publication	Keep last-known-good module cache
Secrets/API keys	Never embed secrets in the NABU binary

Table 10. Permanent modern-host / NABU responsibility split. [S09] [S11]

13.1 Integrated Gateway launcher and dependency preflight

The production integrated entry point is `START_COMMAND_CENTER_GATEWAY.BAT`. It prefers ``py -3``, falls back to ``python``, verifies the installed Python distribution reports ``mido==1.3.3`` before importing/starting the Gateway, and refuses startup with an actionable requirements-music.txt installation command when the requirement is absent or wrong. ``requirements-music.txt`` pins ``mido==1.3.3``. The actual Python executable/version must be captured in the release manifest rather than inferred from a generic Python installation. [S16]

The integrated launcher starts `command_center_gateway.py` with `command_center.ini`. Gateway logs are host-side files governed by the configured ``log_path``; the NABU client has no persistent error-log file. The Store root remains `D:\NABU Internet Adapter\Store`. Modern source failures must preserve last-known-good bounded resources where the module policy allows it.

13.2 Current provider and reduction map

Module	Provider	Current endpoint family	Output	Production note
Atomic Clock	NIST Internet Time Service	time.nist.gov	ncc_time.dat	NIST recommends the generic load-balanced hostname; checked 2026-08-26.

Module	Provider	Current endpoint family	Output	Production note
Weather / location	NWS API + Census ZCTA	api.weather.gov /points, stations, latest observations; Census Gazetteer ZCTA coordinates	ncc_weather.dat / ncc_location.dat / ncc_wxhist.dat	Gateway resolves/reduces; NABU does not ingest JSON.
Weather alerts	NWS Alerts API	api.weather.gov/alerts family	ncc_alert.dat	Bounded severity/count/text; NWS documents JSON-LD/CAP alert services.
Earthquake	USGS	Real-time GeoJSON Summary Feed; current Gateway uses 2.5_week family	ncc_quake.dat	Gateway retains bounded local/global top events.
Space Weather	NOAA/SWPC	services.swpc.noaa.gov JSON products	ncc_space.dat	Gateway reduces current + history arrays.
Satellite / ISS	WhereTheISS	api.wheretheiss.at/v1/satellites/25544	ncc_satellite.dat	ISS NORAD id 25544; bounded SA01.
Airspace / ADS-B	ADSB.lol	/v2/point/{lat}/{lon}/{radius}; production radius 100 NM	ncc_airspace.dat	Gateway retains up to three aircraft and prepares display coordinates.
Music	Mutopia + Mido	Mutopia piece 257 MIDI; host parser mido==1.3.3	ncc_music.dat	Whole-file validate/convert/cache before NABU playback; no playback-time web dependency.

Current implementation source is authoritative for exact URLs used by this release. External service references checked 26 Aug 2026: NIST Internet Time Service; NWS API/Alerts documentation; U.S. Census Gazetteer ZCTA record layout; USGS Earthquake GeoJSON feeds; NOAA/SWPC Data Service; WhereTheISS developer API; ADSB.lol API; Mutopia piece 257; Mido documentation. [S16] [W01]-[W09]

14. Publication transaction

Figure 7. Gateway publication transaction. Partial or invalid modern-side state never replaces a last-valid NABU resource.

The Gateway should acquire, validate, normalize, reduce, encode, checksum, and publish atomically. Store readers should see either the previous complete record or the next complete record - never an in-place partial update. Provider-specific raw documents remain outside the NABU. [S08] [S09]

15. Production resource registry

Resource	Size	Direction	Purpose
ncc_zip.req	17 B	NABU -> Gateway	ZIP sssss zzzz\n request
ncc_time.dat	64 B	Gateway -> NABU	NIST time / sequence / UTC
ncc_weather.dat	64 B	Gateway -> NABU	Current weather for selected ZIP
ncc_location.dat	64 B	Gateway -> NABU	LC01 token/ZIP/city/state

Resource	Size	Direction	Purpose
ncc_wxhist.dat	64 B	Gateway -> NABU	WXH1 rolling Weather history for the accepted request token/ZIP
ncc_alert.dat	64 B	Gateway -> NABU	WA01 bounded weather alert
ncc_quake.dat	64 B	Gateway -> NABU	EQ01 up to four local/global events
ncc_space.dat	64 B	Gateway -> NABU	SW01 current and history
ncc_satellite.dat	64 B	Gateway -> NABU	SA01 ISS state
ncc_airspace.dat	64 B	Gateway -> NABU	AS01 up to three aircraft
ncc_music.dat	512 B	Gateway -> NABU	MU01 music event container; read in 8 x 64 B

Table 11. Current production Store resources. Sizes are derived from current source/build evidence; exact Gateway encoders are authoritative for publication semantics.

16. Shared record rules

- Validate magic/version before interpreting payload.
- Validate length before field access.
- Validate checksum/integrity before committing state.
- Validate token/ZIP/sequence when the resource is selection-specific.
- Use little-endian conversion where the current schema defines multi-byte integers.
- Reject malformed values and preserve the last valid cache.
- Do not parse raw web formats on the Z80.
- Do not mutate the only copy of a record before integrity validation unless the checksum model explicitly accounts for it.

17. Current byte-level fixed-record contracts

17.1 LC01 location record

Offset	Field	Meaning
0-3	'LC01'	Magic/version
4-9	request token	Six decimal characters
10-14	ZIP	Five decimal characters
15	status	0 unresolved / 1 resolved
16	city length	1..16
17	state length	must be 2
18-33	city	Uppercase ASCII, bounded
34-35	state	Two uppercase letters or '--'
36-56	reserved	must be zero
57-58	checksum16	sum of bytes 0..56, little-endian

Offset	Field	Meaning
59-62	'END!'	terminal marker
63	LF	0x0A

Table 12. LC01 contract as implemented by the current NABU validator. [S13]

17.2 EQ01 earthquake record

Offset	Field	Meaning
0-3	'EQ01'	Magic/version
4	sequence	8-bit sequence
5	local count	0..2
6	global count	0..2
7	reserved/header	Gateway-defined
8..	13-byte event slots	mag10, depth, lat100, lon100, age-minutes, 4-char region, event-id
60-61	checksum16	sum of bytes 0..59
62	'E'	terminal marker
63	LF	0x0A

Table 13. EQ01 client-validated structure. [S13]

17.3 SW01 space-weather record

Offset	Field	Meaning
0-3	'SW01'	Magic/version
4	sequence	8-bit
5	Kp x10	e.g. 7 => 0.7
6-7	solar-wind speed	unsigned 16-bit
8	Bt x10	bounded magnetic field metric
9	Bz x10	signed byte
10	F10.7 / flux byte	current reduced flux metric
11	history count	1..12
12-23	Kp history	up to 12 bytes
24-47	speed history	up to 12 little-endian u16 values
48-51	source UTC	u32
52	severity	Gateway-reduced severity
60-61	checksum16	sum 0..59
62	'E'	terminal marker
63	LF	0x0A

Table 14. SW01 client-validated structure. [S13]

17.4 SA01 satellite record

Offset	Field	Meaning
0-3	'SA01'	Magic/version
4	sequence	8-bit
5	state/version	must be 1 in current client
6-7	latitude x100	signed 16-bit
8-9	longitude x100	signed 16-bit
10-11	altitude km	u16
12-13	velocity / 10	u16 then x10 km/h
14-17	timestamp	u32
18-22	identity	5 fixed chars
23-27	NORAD	5 fixed chars
28	visibility	enum
29-30	footprint km	u16
60-61	checksum16	sum 0..59
62	'E'	terminal marker
63	LF	0x0A

Table 15. SA01 client-validated structure. [S13]

17.5 AS01 airspace record

Offset	Field	Meaning
0-3	'AS01'	Magic/version
4	sequence	8-bit
5	state	1..3
6	source	1..2
7	aircraft count	0..3
8..58	3 x 17-byte slots	callsign[6], ICAO[6], x, y, altitude/100, speed/2, heading/2
60-61	checksum16	sum 0..59
62	'E'	terminal marker
63	LF	0x0A

Table 16. AS01 client-validated structure. Current x/y are bounded display coordinates prepared Gateway-side. [S13]

17.6 MU01 music container

Offset	Field	Meaning
0-3	'MU01'	Magic/version
4	format version	1
5	event count	1..82
6	timing model	1 in current client
7	source	1 or 2
8-11	reserved/header	Current encoder/validator-owned bytes; reproduce from exact source
12..	6-byte events	duration, periodB lo/hi, periodC lo/hi, packed B/C volume
504-505	checksum16	sum of bytes 0..503, little-endian
506	reserved/state	Current encoder/validator-owned byte; reproduce from exact source
507-510	'END!'	terminal marker
511	LF	0x0A

Table 17. MU01 client validation contract. [S11] [S13]

17.7 ZIP request contract

Byte(s)	Field	Contract
0-3	ASCII 'ZIP '	Literal request prefix.
4-9	request token	Exactly six decimal digits.
10	' '	Literal separator.
11-15	ZIP	Exactly five decimal digits.
16	LF	0x0A terminator.

Outbound Store request: ncc_zip.req, exactly 17 bytes, no checksum. Matching token/ZIP binds later location/weather/alert state. [S16]

17.8 TIME record contract

Field order	Field	Current validator contract
1	NCC	Magic.
2	1	Protocol version.
3	TIME	Record type.
4	sequence	Six decimal digits; rollback rejected; same sequence is CACHED.
5	UTC	Unsigned decimal epoch value, validated as u32; new-sequence UTC rollback rejected.
6	declared text length	Two decimal digits; must match text length.
7	value	Four decimal digits; <=9999.
8	text	Sanitized ASCII payload.

Field order	Field	Current validator contract
9	checksum	Four hex digits; additive checksum over record body through the separator before checksum.
10	END	Terminal field; line ending stripped for validation.
Physical size	64 bytes	Gateway publishes canonical 64-byte Store record; client reads/validates bounded record.

TIME uses the production NCC|1|TIME delimited envelope. The exact current Gateway encoder and LV2 validator control padding/line-ending details; do not substitute the historical TEST fixture. [S16]

17.9 WX current-weather record contract

Field	Width / rule	Meaning
Magic/version/type	NCC 1 WX	Delimited ASCII envelope.
request token	6 decimal digits	Must match current client request token.
ZIP	5 decimal digits	Must match currently selected ZIP.
temperature	3-character production field	Bounded current temperature representation.
condition	8-character production field	Sanitized bounded condition text.
wind	3-character production field	Bounded wind representation.
pressure	4-character production field	Bounded pressure representation.
source UTC	10 decimal digits	Gateway source/publication time as current encoder defines.
checksum	4 hex digits	Additive checksum through separator preceding checksum field.
end	END + LF within 64-byte canonical record	Client validates exact record length and envelope.

ncc_weather.dat is exactly 64 bytes. Token/ZIP mismatch, malformed envelope, checksum failure, or field-range failure does not replace the last accepted Weather cache. [S16]

17.10 WXH1 weather-history record contract

Offset	Field	Contract
0-3	`WXH1`	Magic/version.
4-9	request token	Six decimal characters.
10-14	ZIP	Five decimal characters.
15	sample count	1..12.
16	reserved/state	Current encoder/validator-owned byte; reproduce from exact source.
17-20	source UTC	Little-endian u32.
21..56	samples	Three bytes per sample, bounded by sample count.

Offset	Field	Contract
57-58	checksum16	Additive sum of bytes 0..56, little-endian.
59-62	`END!`	Terminal marker.
63	LF	0x0A.

ncc_wxhist.dat is exactly 64 bytes, Gateway -> NABU. The client accepts it only for the current Weather selection/token context. [S16]

17.11 WA01 weather-alert record contract

Offset	Field	Contract
0-3	`WA01`	Magic/version.
4-9	request token	Six decimal characters.
10-14	ZIP	Five decimal characters.
15	sequence	8-bit current alert sequence.
16	severity	0..4.
17	count	Bounded alert count.
18	text length	Length of printable alert text.
19-22	source UTC	Little-endian u32.
23..60	text / bounded payload	Printable text only; exact accepted length governed by byte 18.
61-62	checksum16	Additive sum of bytes 0..60, little-endian.
63	LF	0x0A; WA01 has no END marker.

ncc_alert.dat is exactly 64 bytes. A new invalid record cannot replace the last valid alert state. [S16]

PART VI - Native Z80 Client Architecture

18. Program organization

The current native client separates persistent module caches, renderers, input/view state, the 60 Hz timebase, refresh dispatch, AY cue/music ownership, RetroNET reads, and diagnostics. It uses fixed global arrays and compact state rather than dynamic allocation. The main loop services clock, cue audio, scheduler/global refresh, guard checks, then reads one nonblocking key. [S13]

```
for (;;) {
  service_clock();
  service_sound();
  service_scheduler();
  diag_check_guards();
  key = getk();
  if (key == 0) { bounded_delay(); continue; }
  dispatch_key(key);
}
```

19. 60 Hz timebase and refresh

Figure 8. Current timing model: a minimal VDP frame counter supports the local clock, Music, and automatic refresh while manual R remains independent.

The ISR is intentionally minimal: it increments a frame counter. `service_clock()` converts elapsed frames into seconds and accumulates the 60-second automatic-refresh phase. Manual and automatic requests use separate saturating pending counters. `service_global_refresh()` gives a pending manual request priority over automatic work, a `refresh_in_progress` guard prevents overlap, and all transport remains foreground/main-loop work rather than ISR work. A manual request does not reset or postpone the automatic phase. Current source also defines one fixed foreground module-refresh order; because that exact call sequence was not independently recoverable from the documentation corpus used to create this revision, it must be copied verbatim from the final frozen LV2 source into the release manifest rather than inferred here. [S16]

A critical lesson is not to use main-loop iteration counts as real time. The first Music candidate did exactly that and produced grossly stretched notes. The corrected player interprets duration units as 60 Hz frames. The same principle applies to any new timing-sensitive feature.

20. View and control architecture

Key	Raw code	Context	Action
Right / Left / Up / Down	F0 / F1 / F2 / F3	Dashboard	Move selected tile.
Left / Right	F1 / F0	Earthquake, Satellite, Airspace detail	Select bounded target.
ENTER	0D	Dashboard	Open selected module.
ENTER	0D	Target-aware detail	Toggle target lock.
ESC	1B	Global views	Return/back to dashboard.
M / m	ASCII	Global views	Return/back to dashboard.
R / r	ASCII	Global	Queue immediate manual global refresh; does not move auto-refresh deadline.
Z / z	ASCII	Dashboard only	Open exact five-digit ZIP editor.
Backspace / Delete	08 / 7F	ZIP editor	Remove one digit if present; safe at empty.
L / l ; G / g	ASCII	Global	Local / Global mode.
P / p	ASCII	Global	Pause/resume scheduler animation.
X / x	ASCII	Global	Toggle UI sound cues.
B / b	ASCII	Global	Toggle background Music.
N / n	ASCII	Dashboard only	Advance bounded news/status line.
D / d	ASCII	Global	Diagnostics.
H / h	ASCII	Global	Single-page Help.
Q / q	ASCII	Global	Safe Idle.
J / j ; U / u ; E / e	ASCII	Engineering controls	Gate-0 bounded loops 100 / 1000 / 5000.
Left / Right	F1 / F0	Task Manager	Select page 1 / 2 / 3.

Key	Raw code	Context	Action
C / c	ASCII	Any	UNBOUND keyboard key in current production; do not confuse with AY channel C.

Table 18. Current production input map, raw key codes and context precedence from LV2 source. Global keys are consumed before view-specific handling. [S16]

21. Graphics architecture

The owner-frozen GUI uses TMS9918A Mode 2 with a custom 5x7 microfont, fixed safe regions, vector lines, status colors, a persistent command header, six 3x2 dashboard panels, a status/news strip, per-module maximized detail views, and a right-side telemetry rail. Redraw is state-driven: static and dynamic regions are separated so the application does not rebuild the entire display continuously. [S12] [S13]

Logical region	Purpose
Header	NABU COMMAND CENTER, Atomic Clock, Music/Sound, ZIP/location, Local/Global, network status
Dashboard grid	Six owner-frozen module tiles
Ticker/status	Bounded current status/news/alert summary
Maximized plot	Module-specific vector/trend/terrain/orbit visualization
Right telemetry rail	Truthful source/status/selected-target metrics
Footer	Controls and visible BUILD_ID

Table 19. Frozen screen architecture.

22. AY channel ownership

AY resource	v1.0 owner	Rule
Channel A	UI cues / splash melody when splash owns startup	Background Music does not own A during normal application.
Channel B	Background Music voice 1; splash noise/snare during startup	Use safe mixer writes.
Channel C	Background Music voice 2	Unused by splash; normal Music owns as needed.
Register 7	Shared audio + NABU I/O-critical state	Preserve required non-audio bits; never copy an imported player's blanket register-7 writes.
AY I/O ports	Machine I/O	Never treat as arbitrary music payload registers.

Table 20. AY ownership and safety. [S04] [S05] [S11]

PART VII - Version 1.0 Live Modules

Every module follows the same construction template: source provenance -> Gateway acquisition -> normalize/reduce -> bounded resource -> NABU validator -> last-valid cache -> summary renderer -> maximized renderer -> focused host/static tests -> one production build -> Owner MAME acceptance -> freeze -> later integrated physical acceptance. The following sections document the current implementation rather than the older planned

states. [S10] [S12]

Subsystem	Primary source	Store resource	Size	NABU presentation
Atomic Clock	NIST NTP maintenance	ncc_time.dat	64 B	Date/time, local 60 Hz advance, cache/live state
Local Weather	NWS observations	ncc_weather.dat	64 B	Temperature, condition, wind, pressure, history/trends
Weather Alerts	NWS alerts	ncc_alert.dat	64 B	Severity/count/bounded text summary
Earthquake	USGS GeoJSON feed	ncc_quake.dat	64 B	Local/global events, mag/depth/age/region
Space Weather	NOAA/SWPC JSON products	ncc_space.dat	64 B	Kp, wind, magnetic/flux values, 12-sample histories
Satellite / ISS	WhereTheISS	ncc_satellite.dat	64 B	ISS identity, lat/lon, altitude, velocity, visibility
Airspace / ADS-B	ADSB.lol /v2 point	ncc_airspace.dat	64 B	Top 3 callsign/ICAO, plot coords, altitude/speed/heading
Music	Mutopia + Mido conversion	ncc_music.dat	512 B	MU01 two-voice AY event segment

Table 21. Current source/resource registry. Modern service URLs are version-sensitive and must be rechecked before a new release. [S11] [W01] [W02] [W03]

23. Atomic Clock

The Gateway maintains time using NIST and publishes a bounded time record. The NABU validates sequence/UTC state, synchronizes last_utc, resets its frame-derived local phase, and then advances the displayed UTC time locally once per second. Automatic refresh later resynchronizes upstream. A stale-pixel defect was corrected by clearing the exact clock rectangle before redrawing the full 15-character value. The final behavior is live, readable, and refresh-independent.

Ground-up rule: prove the 60 Hz frame counter before adding local epoch advance; prove time-record validation before calling the screen synchronized; and clear only the clock's bounded pixel band on each second.

24. Dynamic ZIP and location

ZIP entry is an explicit bounded request, not a client-side ZIP database. The NABU accepts exactly five digits, builds a 17-byte `ZIP|ssssss|zzzzz|n` request with a six-digit sequence/token, writes it to ncc_zip.req through the existing Store API, marks prior location/weather cache ineligible for the new selection, and waits for matching Gateway publications. The Gateway uses Census ZCTA coordinates and NWS point metadata to derive a practical city/state label; this is not claimed to be canonical USPS addressing.

The LC01 validator requires matching token and ZIP before accepting the city/state result. If resolution fails, the client reports a truthful UNKNOWN/UNRES state rather than inventing locality.

25. Local Weather and Weather Alerts

The Weather Gateway publishes a compact weather record for the current ZIP and a separate NWS alert record. The client validates exact length, envelope, token, ZIP, field count, UTC, checksum/end marker and field ranges before updating its cache. The maximized view presents current temperature, pressure, wind, condition and trends while the

alert subsystem uses a stable bounded summary rather than the abandoned continuously scrolling crawl.

A late regression exposed a concrete stack bug in `draw_weather_detail()`: title-building indices were reused while formatting `metric1`, and an unsafe condition dereferenced `weather_temp[i]` before checking `i<3`. The fix reset independent buffer indices and moved bounds checks before dereference. Focused testing used ZIP 85234 / GILBERT AZ and maximum-width valid weather fields. NCC-SPLASH-260826-LV2 is the corrected 59,501-byte artifact subsequently verified in MAME and on physical NABU.

Weather renderer safety rule	Production requirement
Independent buffers	Each destination begins with independent indices.
Bounds order	Check index limit before source dereference.
Termination	Reserve capacity for NUL and terminate deterministically.
Stack discipline	Prefer bounded locals; do not solve index bugs by simply enlarging stack buffers.
Regression fixture	Exercise current live-style maximum-width values.

Table 22. Weather renderer memory-safety lessons from the final v1.0 correction.

26. Earthquake

The USGS Gateway reduces recent GeoJSON events into an EQ01 record containing up to two local and two global events. The NABU stores magnitude x10, bounded depth, signed latitude/longitude x100, age minutes, a four-character region code and a compact event id. Left/Right cycles targets in the maximized seismic view; ENTER toggles target lock. [S13] [W02]

27. Space Weather

The NOAA/SWPC path publishes current Kp, solar-wind speed, magnetic/flux metrics, history arrays, source time and severity. The accepted display labels Kp history, solar-wind history and a fixed Kp5 storm threshold so the graph is interpretable rather than merely decorative. NOAA's current data service exposes JSON products including planetary K-index and related space-weather products. [S12] [W03]

28. Satellite / ISS

The v1.0 satellite service tracks ISS/NORAD 25544. Gateway-side modern Internet handling is reduced into SA01; the NABU validates exact length, checksum, state, identity, position, altitude, velocity, timestamp, visibility and footprint fields. The owner-approved orbital/vector presentation remains intact while mock status labels were removed in favor of live values. The current renderer clears only its bounded right rail before redrawing updated live telemetry.

29. Airspace / ADS-B

Airspace uses the ADSB.lol point API centered on the selected ZIP/location, with a 100 NM search radius in the production design. The Gateway retains at most three aircraft, computes range/bearing and display projection, and publishes callsign, ICAO, display x/y, altitude, speed and heading in AS01. The NABU never parses the provider's large aircraft JSON. ADSB.lol currently documents ``v2/point/{lat}/{lon}/{radius}`` and permits radii up to 250 NM; its current public API/data license is ODbL 1.0, so release attribution/terms must remain part of publication review. [W01]

The owner-frozen visualization is a perspective vector volume, not a top-down radar. The accepted LV3 restored the original vanishing-point design, expanded the floor to the safe viewport, added bounded marker decluttering and Gateway-side scale/projection, and kept the right telemetry rail unchanged. This is an important design-governance case: a technically plausible top-down redesign was rejected because the approved mockup is owner-frozen.

Figure 9. Owner-captured physical NABU evidence: live Airspace / ADS-B perspective display on NCC-SPLASH-260826-LV2.

30. Integrated feature acceptance

Figure 10. Owner-reported integrated acceptance matrix for NCC-SPLASH-260826-LV2. MAME and physical-NABU gates are distinct. Included photographs directly show selected screens; the matrix also reflects the Owner's broader integrated-operation report and must not be read as photographic proof of every plotted point.

PART VIII - Task Manager, Diagnostics, Help, and Resource Accounting

31. NABU System / Task Manager

The Task Manager is a truthful application/system diagnostic view, not a modern OS process monitor. It intentionally distinguishes direct changing runtime state from calculated values, linker/map-derived values, documented machine constants, and unavailable measurements. Unsupported CPU utilization, temperature, voltage, load, and 'free RAM' values are not fabricated.

Classification	Meaning	Examples in current UI
LIVE	Direct current application/runtime state	screen/view, ZIP, draw stages/counters, dirty mask, scheduler pause, module, last I/O, time sequence
CALC	Reproducible calculation from current allocations/state	net buffers, font tables, Mode-2 VRAM allocation, sprite allocation
MAP	Current linker/map-derived value	DATA+BSS section total
CONST	Documented NABU hardware fact	Z80A, ~3.58 MHz, 64K RAM, TMS9918A, 16K VRAM, AY-3-8910
UNAVAILABLE	No truthful supported source in current build	user data, undefined link-gap metric, unsupported globe fields

Table 23. Task Manager truth classifications.

31.1 Task Manager field provenance ledger

Displayed area	Class	Source/formula	Release meaning
Hardware identity	CONST	Documented NABU platform facts	Z80A, ~3.58 MHz, 64K CPU RAM, TMS9918A, 16K VRAM, AY-3-8910.
MAP D+B	MAP	Current linker/map DATA + BSS total	1104 B in the reviewed Task Manager build; scoped to linker DATA+BSS, not total/free RAM.
USER DATA	UNAVAILABLE	No truthful current source	Displayed UNAVAILABLE.
NET BUF	CALC	sizeof(record_buffer)+sizeof(par se_buffer)	Calculated from current fixed buffers.
FONT C	CALC	10*7 + 26*7 + 9*7	315 bytes.
LINK GAP	UNAVAILABLE	No defined truthful metric	Displayed UNAVAILABLE; earlier SEE MAP label retired.

Displayed area	Class	Source/formula	Release meaning
VRAM	CALC	Mode-2 allocation calculation	13056 bytes in current Task Manager display.
Right-rail provenance	CONST/MAP	Renderer labels	Explicitly distinguishes hardware constants from map-derived data.
Runtime view/module/counters	LIVE	draw_task_live()/draw_task_internals() current state	Direct application state/counters; exact field reset semantics remain source-defined and should be copied to release manifest at freeze.

This ledger records fields independently verified by Codex against the current source/map. The 646-byte FC7A-to-FF00 quantity is address-space/static-to-stack margin and must not be labeled free RAM. [S16]

The final Task Manager correction removed stale hardcoded memory values, changed ambiguous `DATA/BSS` to scoped `MAP D+B`, replaced `LINK GAP / SEE MAP` with `UNAVAILABLE`, removed a stale legacy build literal from Diagnostics, corrected mislabeled counters, and retained the shared current BUILD_ID. NCC-TASK-260826-LV2 built as 58,384 byte-identical bytes with SHA-256

1B0C00C9B9BBF332444D30641737455005B3B97B17E01DE47AB508BE932B5C6C before Music and splash were layered on top.

Figure 11. Owner-captured physical NABU evidence: Task Manager hardware/resource page in the current integrated build.

32. Diagnostics and flight-recorder note

The source retains diagnostic event IDs, cumulative/runtime counters, guard bytes, visible diagnostic status and compile-time ring declarations. In the splash-era LV2 production source, `NCC\_DIAG\_FLIGHT\_RECORDER` defaults to 0: the ring storage is compiled out and `diag\_log()` is a no-op, while guards/counters/checkpoints used by visible diagnostics and debugger inspection remain. There is no persistent NABU-side file/error log. Gateway logging is a separate modern-host function controlled by its configured `log\_path`. Do not describe the disabled ring as built-in error logging. [S16]

The correct debugging capture for a stop/hang remains: BUILD_ID, last visible checkpoint, PC, SP, surrounding disassembly, registers/flags, map/listing symbol, interrupt state, and whether MAME is paused. [S01] [S08]

33. Basic Help and the v1.0 resource gate

A basic single-page VIEW_HELP already exists and is entered with H. It lists current verified controls and exits with ESC or M. A requested 6-8 page DOS-style About/Help manual was intentionally not implemented after the current link map showed only 646 bytes between BSS_END \$FC7A and the verified initial stack at \$FF00. No verified Page Up/Page Down key codes were present in the project/headers; Left/Right were already verified. The correct engineering decision was to stop rather than consume unsafe stack headroom or strip production features.

Figure 12. Current memory headroom explains why a dense multi-page Help system is a v1.0 resource decision rather than a trivial text addition.

PART IX - Music and Startup Splash

34. Music design evolution

The Music white paper originally specified a global header service with reuse-first host conversion, multiple source candidates, genre controls, automatic source failover, and AY B/C playback while reserving A for cues. That paper

explicitly warned against reinventing a player or file format until NabuTracker, Nth Pong Wars/CHIPNSFX, Arkos and z88dk lineages had been inspected. [S11]

The actual v1.0 implementation narrowed the policy: Mutoxia became the production source, the keyboard C key remains unbound; AY channel C remains Music voice 2, and a single curated stream was chosen to avoid adding category/catalog machinery solely for genre switching. The Gateway downloads a complete MIDI, validates it, converts it with Mido 1.3.3 into a fixed MU01 cache, and the NABU loads the complete 512-byte resource in eight already-proven 64-byte reads before playback. There are no playback-time transport calls.

Figure 13. Final v1.0 Music data path. The Internet source is a catalog/acquisition source, not a continuous audio stream.

35. Music playback defects and final model

Defect found in early Music candidate	Evidence-supported correction
Mido delta ticks treated without proper tempo conversion	Use ticks_per_beat, default 500,000 us/beat and set_tempo changes; quantize to 60 Hz frames.
Duration multiplied by main-loop iteration count	One MU01 duration unit = one 60 Hz frame; bounded catch-up consumes elapsed frames.
Voice identity tracked too weakly	Track (MIDI channel, note) identity and emit complete B/C state snapshots.
82-event limit silently clipped a phrase	Reject unsafe truncation; choose a clean all-voices-silent boundary within the fixed container.
AY periods rounded differently from NabuTracker oracle	Use verified truncation/table behavior for representative MIDI notes.
Old structurally valid cache could bypass new converter	Add converter identity/version gate and reconvert cached complete MIDI.

Table 24. Music defects corrected before acceptance.

The accepted v1.0 work became J. S. Bach's public-domain 'Aria Bist Du Bei Mir' from Mutoxia. The bounded production segment contains 81 events, runs 2,217 frames / 36.95 seconds, ends at a real all-voices-silent boundary, and loops cleanly. A reported 'hang' around 30 seconds was later reproduced as the musical content itself: a silent boundary at 30.00 seconds followed by a sparse monophonic tail. A deterministic three-loop simulation showed no index, duration, catch-up, or stale-voice corruption.

This implementation is deliberately constrained and is not the maximum musical capability of the NABU/AY. Richer multi-instrument tracker-style playback is a future enhancement, not a v1.0 stability requirement.

36. Startup splash

The splash is a startup presentation only. It uses the existing Mode-2 renderer and 5x7 font primitives to reproduce the owner-approved frame, command-center title hierarchy, NCC reticle, Version 1.0 identity, creator credit, date and 'PRESS ANY KEY'. It appears after the minimal frame ISR is installed and before the normal dashboard is drawn. [S14]

The final startup path uses a 600-frame / 10-second maximum. Any key immediately silences splash audio, drains the dismissal event, and enters the existing dashboard so the key cannot trigger an application command. If no key is pressed, the splash times out automatically. The music is a fixed embedded eight-bar public-domain 'Yankee Doodle' phrase: 32 note indices, seven AY periods, exactly 480 frames / 8 seconds, Channel A melody plus Channel B noise/snare, then silence until the splash exits.

Figure 14. Owner-captured physical NABU evidence: final v1.0 splash running on the real machine.

37. Splash memory recovery

The first splash link crossed the Z80 64K boundary. The correction remained splash-local: rendering and tune storage were compressed and the already-disabled diagnostic ring was compiled out, avoiding removal of live modules or owner-approved GUI behavior. The resulting current build fits, but the measured 646-byte static-to-stack margin is now an explicit release constraint. This is why later About/Help expansion stopped at the resource gate.

PART X - Internet Adapter, MAME, Debugger, and Physical NABU

38. NABU Internet Adapter runtime roles

Role	Current verified usage
Program loading	Local HomeBrew folder containing 000001.NABU; MAME talks to Adapter on TCP 5816.
Runtime data	RetroNET Store at D:\NABU Internet Adapter\Store; associated TCP service 5815 in verified environment.
Cache/listener handling	After replacing program 000001.NABU, invalidate/restart the program-loader listener/cache according to the installed Adapter behavior.
Application data	Do not treat TCP 5816 as a general runtime application socket.
Physical NABU	Use documented serial/RS-422 mode for the exact hardware; do not infer settings from MAME TCP behavior.

Table 25. Internet Adapter responsibilities and transport separation. [S08] [S12]

39. MAME 0.250 commands

Normal acceptance:

```
mess nabupc -window -kbd nabu_hle -hcca null_modem -bitb socket.127.0.0.1:5816
```

Debugger:

```
mess nabupc -debug -window -kbd nabu_hle -hcca null_modem -bitb socket.127.0.0.1:5816
```

These are version-specific commands from the verified MAME 0.250 environment. The verified executable path is ``<MAME_ROOT> NABU 9938\mess.exe``; run the command from that directory or prepend the executable path. Normal MAME is the first acceptance environment; use ``-debug`` only when execution evidence is needed. In the tested debugger, F5 is the locally verified resume action. [S08] [S16]

39.1 Exact MAME / Adapter startup order

1. Start the integrated Gateway with START_COMMAND_CENTER_GATEWAY.BAT and confirm required Store resources are publishing under D:\NABU Internet Adapter\Store.
2. Open NABU Internet Adapter. In Settings -> Source, enable the verified local HomeBrew folder and point it at the exact current client directory containing 000001.NABU.
3. Confirm the Adapter's RetroNET Store root is D:\NABU Internet Adapter\Store and the current MAME-era configuration associates runtime Store service with TCP 5815.
4. Click Start TCP for the program loader and confirm the Adapter reports the TCP server on 5816. After replacing 000001.NABU, stop/start this loader to invalidate program/PAK cache before relaunch.
5. Launch MAME 0.250 using the exact command in Section 39. Verify the visible BUILD_ID before evaluating any behavior.

6. Do not restart the Adapter merely to make a new Store telemetry record visible; Store freshness and program-loader cache invalidation are separate concerns.

MAME/Adapter configuration is software-reconstruction evidence. It does not specify the separate physical serial/RS-422 path. [S08] [S16]

40. Standard MAME acceptance sequence

1. Start or verify only the integrated Gateway components required by the test.
2. Start the Internet Adapter once and confirm the correct local program folder and TCP listener.
3. Confirm the intended Store directory and expected Gateway publications.
4. Launch one MAME instance; verify the visible BUILD_ID before evaluating behavior.
5. Exercise one feature at a time. For ZIP tests, enter five digits slowly and wait for Gateway publications.
6. Confirm automatic refresh separately from manual R.
7. Confirm summary view and maximized view for each module.
8. Exercise Music/Sound and return/navigation behavior.
9. Capture screenshots only after the display settles.
10. Use the debugger only when the observed failure requires PC/SP/register/symbol evidence.

41. Physical NABU boundary

Physical NABU is the highest runtime authority, but it is a separate evidence gate. MAME program loading over localhost proves nothing about the electrical interface, USB-to-RS-422 device, COM port, cable pinout, driver, or Adapter physical-mode settings. The Owner has verified the current LV2 binary on real hardware, but those interconnect details are not yet frozen into a reproducible deployment manifest. Therefore this document is a complete software reconstruction and verified-operation record; third-party physical interconnect reconstruction additionally requires the Owner-verified hardware deployment manifest defined below. No missing physical value is guessed. [S01] [S08] [S16]

Physical evidence item	Required before a reproducible deployment
NABU	Exact model/revision and relevant board evidence
Host interface	Exact USB/RS-422 manufacturer/model/chipset/driver
Windows device	Observed COM port and driver state
Electrical mode	Verified compatibility from device + NABU documentation
Cable	Verified connector/pinout/continuity
Adapter	Exact physical mode and settings for installed release
Software	Exact BUILD_ID, artifact hash and source commit/tag

Table 26. Physical deployment evidence record. Never guess missing values.

41.1 Physical deployment manifest required for third-party replication

Manifest field	Current status	Required release record
NABU model/revision	UNVERIFIED in current paper	Owner must record exact machine/board evidence.
USB/RS-422 interface model/chipset	UNVERIFIED in current paper	Record manufacturer/model/chipset/driver.

Manifest field	Current status	Required release record
Windows COM port	UNVERIFIED in current paper	Record observed COM port at acceptance.
Electrical/serial mode	UNVERIFIED in current paper	Record exact Adapter/interface settings from matching documentation.
Cable/pinout/continuity	UNVERIFIED in current paper	Attach verified wiring/pinout/continuity record; do not infer.
Internet Adapter physical mode	UNVERIFIED in current paper	Record installed release, selected mode and settings.
Power/connect sequence	UNVERIFIED in current paper	Record owner-verified safe procedure.
Accepted software	VERIFIED	NCC-SPLASH-260826-LV2; SHA-256 5D27 DFBB3ED8DCD96AF828CECC7550EF546 B07C10509E20A71FECFCF0CC89087.

This table deliberately preserves unknown physical details as UNVERIFIED rather than converting owner knowledge into guessed wiring/settings. [S16]

42. Current physical v1.0 evidence

The Owner loaded and exercised NCC-SPLASH-260826-LV2 on physical NABU hardware after the maximized-Weather correction. Photographic evidence shows the splash, Task Manager pages, Airspace, Space Weather, Satellite/ISS, Earthquake and normal Command Center operation. The Owner reported that the integrated system worked on hardware. This establishes physical transfer/execution and GUI/runtime operation for the exact owner-accepted LV2 release-candidate bytes. Repository status is separate: LV2 is an unfrozen working-tree baseline with no current commit/tag. A final repository freeze/tag remains pending Owner authorization. [S15] [S16]

Figure 15. Release evidence gates used throughout the project.

PART XI - Reliability, Troubleshooting, and Defect Case Studies

43. Failure isolation philosophy

The project repeatedly demonstrated why application bugs, rendering bugs, transport bugs, provider failures, Adapter state and emulator state must not be collapsed into one theory. Diagnose the layer indicated by evidence. If a Gateway continues publishing after a NABU crash, that does not prove the transport caused the crash. If the Adapter reports end-of-stream after a client stack overwrite, the Adapter message may be a consequence rather than the cause.

Observed symptom	First evidence to collect
Wrong screen / stale behavior	Visible BUILD_ID; exact artifact hash
No live data	Gateway publication log; Store file bytes; Adapter state
Short/failed read	Returned byte count; handle/open result; transport counters
Screen corruption	Recent renderer/local-buffer changes; PC/SP; guard state; map/listing
Soft reset	BUILD_ID, last view/checkpoint, Adapter/Gateway logs, debugger capture if reproducible
Music drone/tempo error	Exact MU01 bytes, event timeline, 60 Hz duration state, AY period/volume state
Adapter end-of-stream	Determine whether client/session died first; do not assign direction without evidence

Table 27. Layered troubleshooting evidence.

44. Major engineering defect case studies

44.1 Transport overrun

Retired design: delayed HCCA receive service -> UART overrun -> 63/64 -> indefinite wait. Production action: rebase to Golden Transport rather than tuning speculative delays.

44.2 Stale redraw mistaken for networking

An early T1 screen redraw continuously, visually resembling repeated network activity. Source review showed R still owned the actual Store refresh. The correction changed redraw behavior only and left networking untouched. [S08]

44.3 Weather stack corruption

A maximized-Weather formatter reused title indices for metric1 and dereferenced before checking a bound. The resulting stack overwrite produced severe TMS9918 display corruption and could terminate the client session. Source inspection, not transport speculation, found the bug.

44.4 Music timing/content distinction

Music initially sounded like long drones due to concrete timebase errors. After those were fixed, a later 'hang' at about 30 seconds was shown by exact three-loop simulation to be sparse encoded content, not state-machine corruption. The lesson is to reproduce the exact state before changing architecture.

44.5 Splash 64K overflow

The first splash implementation crossed the 64K link boundary. Only splash/diagnostic storage was reduced; no live telemetry module or owner-frozen UI subsystem was stripped to make room.

45. Endurance and transport qualification

The current source retains Gate-0 qualification commands that can alternate TIME and Weather resources for bounded read loops of 100, 1,000 or 5,000 operations while continuing clock, sound, scheduler and input service. These are engineering controls, not normal user functions. The original Golden foundation's 5,000/5,000 result remains the closed transport qualification milestone. [S12] [S13]

PART XII - Release Engineering, Freeze Discipline, and Publication

46. Milestone ledger

Build	Bytes	SHA-256	Commit	Tag	MAME	Physical	Notes
HELLO DEREK v4	7,948	UNVERIFIED in Codex review	UNVERIFIED	UNVERIFIED	historical VERIFIED	UNVERIFIED	Frozen practical baseline; full historical SHA exists in earlier corpus but was not re-verified in this audit.

Build	Bytes	SHA-256	Commit	Tag	MAME	Physical	Notes
NCC-TP-2608 07-T1-02	12,317	5DD4280281E 7C9C2FAAC8 A0A6445C0F3 91AD54A9451 2C9021CE588 38C525A32A	b5fa2483ea9c 548a5a3212c8 681b525dd4e e4029	historical transport tag lineage; exact tag association should be checked at release	VERIFIED	NOT TESTED / historical	Changed-reco rd Store proof; full digest from preserved engineering paper.
NCC-TF-2608 13-GOLDEN-0 1	UNVERIFIED	FA075985BB9 8DCAE8A442 8F12FABD071 39A9C4F6627 F3622D986A5 626CDD4ABD	bd5cb00692b5 67c6095b503d 4c94f4230fda2 33f	ncc-transport-f oundation-v1- mame-verified	VERIFIED	UNVERIFIED	5,000/5,000 Golden transport milestone.
NCC-VS-2608 09-P2D-A04C 3R2	32,165	UNVERIFIED in Codex review	UNVERIFIED	ncc-phase2d-s hell-v0.9-mam e-verified exists	VERIFIED	UNVERIFIED	GUI shell.
NCC-AC-2608 11-P3A-01AR 1	32,201	UNVERIFIED in Codex review	UNVERIFIED	ncc-phase3a-0 1a-atomic-cloc k-v0.1-mame- verified exists	VERIFIED	UNVERIFIED	Atomic Clock.
NCC-SW-260 820-LV2	48,259	UNVERIFIED in Codex review	7a1b25c00fb9 9b2b70bb84b ed0bd7f4aa72 d27e6	ncc-space-we ather-readable -v1-mame-veri fied	VERIFIED	UNVERIFIED	Tag proves source milestone; artifact digest needs full release manifest.
NCC-SA-2608 20-LV1	51,421	UNVERIFIED in Codex review	UNVERIFIED	UNVERIFIED	VERIFIED per historical record	VERIFIED per historical record	Satellite-era physical baseline.
NCC-ADSB-2 60826-LV3	58,337	UNVERIFIED in Codex review	a2f2e08855fc9 1fd84869d14f eb0c42fea55d a7a	ncc-page1-zip- clock-satellite- refresh-airspa ce-v1-mame-v erified	OWNER VERIFIED	UNVERIFIED for exact ADS-B artifact	Broader Page-1 freeze; do not infer exact artifact/tag equivalence beyond evidence.
NCC-TASK-26 0826-LV2	58,384	1B0C00C9B9 BBF332444D3 06417374550 05B3B97B17E 01DE47AB508 BE932B5C6C	dd1bad5e669 9fd0ddba01fd dac384b74192 70b22	ncc-command- center-pre-mu sic-v1-mame-v erified	OWNER VERIFIED	NOT TESTED for exact pre-Music freeze	Last explicitly committed/tag ged full application baseline.
NCC-MUSIC-2 60826-LV2	60,016	UNVERIFIED in Codex review	UNVERIFIED	NONE	OWNER VERIFIED per development evidence	UNVERIFIED	Current working-tree Music lineage.
NCC-SPLASH -260826-LV2	59,501	5D27DFBB3E D8DCD96AF8 28CECC7550 EF546B07C10 509E20A71FE CFCF0CC890 87	NONE / working tree	NONE	OWNER VERIFIED	OWNER VERIFIED	Current exact .bin/.NABU byte-identical; owner-accepte d but unfrozen.

Table 28. Authoritative current milestone ledger after Codex audit. UNVERIFIED means the exact field was not recovered strongly enough for release identity; no abbreviated digest is treated as authoritative. [S16]

47. Pre-Music freeze

Item	Value
Commit	dd1bad5e6699fd0ddba01fddac384b7419270b22
Annotated tag	ncc-command-center-pre-music-v1-mame-verified
BUILD_ID	NCC-TASK-260826-LV2
SHA-256	1B0C00C9B9BBF332444D30641737455005B3B97B17E01DE47A B508BE932B5C6C
MAME	Owner accepted
Physical	Not tested for that exact pre-Music freeze

Table 29. Last explicitly recorded full application freeze before Music/splash.

47.1 Current LV2 Git state

Field	Current audited state
Branch	phase-page1-live-satellite-airspace-v1
HEAD	dd1bad5e6699fd0ddba01fddac384b7419270b22
HEAD tag	ncc-command-center-pre-music-v1-mame-verified
LV2 commit/tag	NONE
Working tree	Modified client source/artifacts and Gateway integration files; untracked Music/tests/evidence and other project files reported by Codex audit.
Runtime status	NCC-SPLASH-260826-LV2 owner-verified in MAME and on physical NABU.
Freeze status	OWNER-ACCEPTED, UNFROZEN WORKING-TREE RELEASE CANDIDATE.

Runtime acceptance does not imply repository freeze. This state must remain explicit until Owner-authorized final commit/tag/manifest work occurs. [S16]

48. Final freeze procedure for the current release candidate

1. Confirm source BUILD_ID is NCC-SPLASH-260826-LV2 and artifact SHA-256 is 5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087.
2. Confirm 000001.bin and 000001.NABU are 59,501 bytes and byte-identical.
3. Confirm current MAME and physical-NABU acceptance evidence is associated with that exact artifact.
4. Record map/listing/resource state including BSS_END \$FC7A, initial SP \$FF00 and the 646-byte margin.
5. Update PHASE_STATE / RELEASE_MANIFEST / VERSION / BUILD_STATE / API_PROVENANCE / acceptance records as applicable without changing production behavior.
6. Commit only the accepted files and generated evidence required by repository policy.
7. Create an annotated release-candidate/final tag only after Owner authorization.

8. Do not push a remote unless explicitly authorized.
9. Package the exact tagged binary; do not build a special distribution-only binary.

49. Open-source/publication package

Package component	Required content
Native client	Source, BUILD.BAT/CLEAN.BAT, exact .bin/.NABU, map/sym/listings as release evidence
Gateway	Integrated source, requirements, config example, launcher/preflight, focused tests
Protocols	Resource registry, byte-level schemas, checksum rules, source registry
Documentation	README, VERSION, BUILD_STATE, API_PROVENANCE, ACCEPTANCE_TESTS, changelog, known limitations
Integrity	SHA-256 manifest and release commit/tag
Licensing	Project license plus third-party/content attribution and per-source policy notes
Evidence	Selected MAME screenshots and physical-NABU photos

Table 30. Recommended v1.0 release bundle. [S10]

50. Version 2.0 boundary

Version 2.0 remains downstream of the v1.0 freeze. Candidate work includes dashboard paging, a second six-module page, expanded diagnostics/help, and a hardened Pi-class Gateway appliance. Page switching must never create a second networking engine; hidden modules retain cached state under the same scheduler. The exact v2 module list remains Owner-controlled. [S10]

PART XIII - Ground-Up Replication Procedure

51. Stage A - Establish the host

1. Create or verify <NABU_WORKSPACE> and <NABU_WORKSPACE>.
2. Restore z88dk at the verified Tools\z88dk path. Confirm bin\zcc.exe and lib\config exist.
3. Restore MAME/MESS under the verified MESS NABU 9938 directory.
4. Install/verify the NABU Internet Adapter. Record exact installed version and executable before version-sensitive work.
5. Install/verify Git; keep the repository local-only unless the Owner authorizes a remote.
6. Install/verify Python 3 for the integrated Gateway. Use START_COMMAND_CENTER_GATEWAY.BAT as the production entry point: it prefers `py -3`, falls back to `python`, and refuses startup unless the installed distribution reports exactly mido==1.3.3. `requirements-music.txt` is the pinned dependency source; record the actual Python executable/version in the release manifest.
7. Create/verify D:\NABU Internet Adapter\Store.

52. Stage B - Prove the native toolchain

1. Restore the frozen Hello Derek source and BUILD.BAT.

2. Run BUILD.BAT without editing target flags.
3. Confirm Z: mapping, zcc +nabu invocation, successful compile/link, 000001.bin and 000001.NABU.
4. Hash the artifacts; compare to the frozen 7,948-byte hash only when reproducing the exact same source/tool state.
5. Point the Adapter's local HomeBrew source to the Hello Derek directory, confirm Store/program-loader configuration, start the TCP program loader on 5816, launch MAME 0.250 using the exact Section 39 command, and verify keyboard-first READY behavior before debugging Command Center.
6. Only after this passes move to Command Center-specific source.

53. Stage C - Reconstruct the Golden transport

1. Restore the Command Center repository and preserved transport-foundation tag.
2. Inspect installed z88dk/NABU-LIB declarations rather than copying tutorial APIs.
3. Implement/restore the bounded Store open-read-close path and protected read wrapper.
4. Prove exact 64-byte valid reads first.
5. Run focused missing/truncated/integrity/recovery tests appropriate to the preserved foundation.
6. Run the bounded 5,000-read qualification before treating transport as closed.
7. Freeze transport independently before adding live modules.

54. Stage D - Build the graphical shell

1. Initialize TMS9918A Mode 2 with the verified existing primitives.
2. Install the minimal frame ISR that only counts frames.
3. Implement the custom 5x7 microfont and safe 256x192 regions.
4. Create the persistent header, six-panel dashboard, news/status strip, maximized frame and right telemetry rail.
5. Implement Arrow/ENTER/ESC navigation, BUILD_ID footer, Sound, Diagnostics, Help and Safe Idle.
6. Prove no continuous full-screen redraw loop.

55. Stage E - Add live services in dependency order

1. Atomic Clock and 60 Hz local advance.
2. ZIP request + location profile.
3. Local Weather, history/trends and stable Weather Alert summary.
4. Earthquake local/global cache and target selection.
5. Space Weather current + history traces.
6. Satellite / ISS live SA01 cache.
7. Airspace / ADS-B live AS01 cache and owner-frozen 3D perspective renderer.
8. Task Manager truth audit.
9. Music whole-file acquisition/conversion/cache and bounded 60 Hz AY playback.
10. Startup splash and fixed embedded march.
11. Integrated regression in MAME, then physical NABU using the same exact accepted binary.

Figure 16. Ground-up system construction sequence. Do not jump directly to complex modules before proving the underlying layer.

56. Acceptance gates for every subsystem

Gate	Pass condition
Design	Scope, controls, source, record and failure policy documented
Source/static	Only authorized files changed; focused host/static tests pass
Compile	Target compiler passes
Link	Linker passes; resource map inspected
Artifacts	Expected .bin/.NABU exist and identity/hash recorded
MAME	Owner verifies exact BUILD_ID and requested behavior
Freeze	Accepted bytes committed/tagged if authorized
Physical	Exact frozen bytes run on the real NABU
Release	Manifest, provenance, licenses, docs and known limitations complete

Table 31. Repeatable subsystem gate model.

APPENDICES

Appendix A - Current production state ledger

Field	Value
Repository	<publication-repository>
Active client	client\phase3a_live_local_weather_v0_1\NABU_COMMAND_CENT ER_LIVE_LOCAL_WEATHER_v0_1
Gateway	gateway\phase3a_live_local_weather_v0_1\NABU_COMMAND_CE NTER_LOCAL_WEATHER_GATEWAY_v0_1
Build	NCC-SPLASH-260826-LV2
Artifact size	59,501 bytes each
SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A7 1FECFCF0CC89087
MAME	OWNER VERIFIED
Physical NABU	OWNER VERIFIED
Current final freeze/tag	NONE for NCC-SPLASH-260826-LV2; Owner-authorized final freeze/tag pending
BSS_END / SP	\$FC7A / \$FF00
Static-to-stack margin	646 bytes
Help expansion	Deferred at resource gate; single-page H Help remains
Repository state	Owner-accepted LV2 working tree; branch phase-page1-live-satellite-airspace-v1; HEAD dd1bad5e...; LV2 uncommitted/untagged at Codex audit

Table 32. Current integrated v1.0 release-candidate state.

Appendix B - Exact current user-control reference

Key	Raw code	Context	Action
Right / Left / Up / Down	F0 / F1 / F2 / F3	Dashboard	Move selected tile
Left / Right	F1 / F0	Earthquake, Satellite, Airspace detail	Select bounded target
ENTER	0D	Dashboard	Open selected module
ENTER	0D	Target-aware detail	Toggle target lock
ESC or M	1B / ASCII M,m	Global views	Return to dashboard/back
R	ASCII R,r	Global	Queue immediate manual global refresh; does not move automatic deadline
Z	ASCII Z,z	Dashboard only	Open exact five-digit ZIP editor
Backspace / Delete	08 / 7F	ZIP editor	Delete one digit when present
L / G	ASCII L,l / G,g	Global	Local / Global mode
P	ASCII P,p	Global	Pause/resume scheduler animation
X	ASCII X,x	Global	Toggle UI sound cues
B	ASCII B,b	Global	Toggle background Music
N	ASCII N,n	Dashboard only	Advance bounded news/status line
D	ASCII D,d	Global	Open Diagnostics
H	ASCII H,h	Global	Open single-page Help
Q	ASCII Q,q	Global	Enter Safe Idle
J / U / E	ASCII J,j / U,u / E,e	Engineering/global	Gate-0 bounded loops 100 / 1000 / 5000
Left / Right	F1 / F0	Task Manager	Page 1 / 2 / 3
C	ASCII C,c	Any	No assigned keyboard action; do not confuse with AY channel C

Table 33. Operator control reference derived from current client source. [S13]

Appendix C - Current status vocabulary

Status	User meaning
LIVE	New/current valid state accepted
CACHE / CACHED	Valid previously accepted state retained
STALE	Valid state too old by the module's freshness policy
OFFLINE / OFF	Provider/transport not currently supplying a valid new record
INVALID	A record or state failed validation

Status	User meaning
UNAVAILABLE	No truthful source/value exists for that field in this build
NOT CFG	Not configured

Table 34. Truthful status semantics used by the native UI.

Appendix D - Current fixed record summary

Record	Bytes	Magic / envelope	Checksum	End
ZIP request	17	ASCII ZIP token ZIP	none	LF
TIME	64	NCC 1 TIME	ASCII additive checksum through pre-checksum separator	END + line ending / canonical 64 B
Weather	64	NCC 1 WX	ASCII additive checksum through pre-checksum separator	END + LF / canonical 64 B
Location	64	LC01	sum 0..56 -> 57-58 LE	END! + LF
Weather history	64	WXH1	sum 0..56 -> 57-58 LE	END! + LF
Weather alert	64	WA01	sum 0..60 -> 61-62 LE	LF only; no END marker
Earthquake	64	EQ01	sum 0..59 -> 60-61 LE	E + LF
Space Weather	64	SW01	sum 0..59 -> 60-61 LE	E + LF
Satellite	64	SA01	sum 0..59 -> 60-61 LE	E + LF
Airspace	64	AS01	sum 0..59 -> 60-61 LE	E + LF
Music	512	MU01	sum 0..503 -> 504-505 LE	END! + LF

Table 35. Current complete Store/request record family. Detailed contracts are in Section 17; exact current encoder/validator source remains authoritative. [S16]

Appendix E - Reference registry

ID	Source	Use
[S01]	Agent_Instructions_SuperPrompt.txt	Primary operating doctrine; evidence hierarchy; owner-frozen design; runtime/network/physical gates.
[S02]	CORPUS_INVENTORY.txt	Authoritative corpus map and inspection-state rules.
[S03]	Nabu_PC_Technical_Specifications.pdf, Spec. 50-90020490, 8 Jun 1984	Original NABU application-programming technical reference: IOS, TMS9918A, keyboard/tasking/network context.
[S04]	Nabu_Computer_Technical_Manual_by_MJ P-compressed.pdf	NABU hardware reference: Z80A clock, 64K RAM, TMS9918A/16K VRAM, AY-3-8910, interrupts, interfaces.
[S05]	Hello_Derek_EXAMPLE_main.c	Frozen practical C/API baseline: getk(), console, AY direct access, sprite bring-up.

ID	Source	Use
[S06]	EXAMPLE_Directories_and_Programs.txt + Hello_Derek_EXAMPLE_BUILD.BAT	Verified Windows paths and +nabu build pattern.
[S07]	Suite_EXAMPLE_main.c	Structured diagnostic/menu/input baseline.
[S08]	NABU_COMMAND_CENTER_Engineering_White_Paper_v1.0.docx, 7 Aug 2026	Initial full architecture, transport proof, MAME/debugger, release and ground-up replication record.
[S09]	NABU_Command_Center_Engineering_White_Paper_2026_Final.docx, 14 Aug 2026	Golden Transport, TIME/Weather, physical regression, GUI rules and production sequence.
[S10]	NABU_COMMAND_CENTER_Engineering_White_Paper_Roadmap_2026-08-25.docx	v1.0/v2.0 release boundary, module/task/music qualification and publication roadmap.
[S11]	NABU_Command_Center_Music_Stream_Module_Engineering_White_Paper_v1.0.pdf	Reuse-first Music design, AY ownership, timing, source policy and validation gates.
[S12]	NABU_COMMAND_CENTER_CODEX_HANDOFF_CURRENT_2026-08-25.txt + subsequent 26 Aug build reports	Current integrated architecture, milestone evidence, final Airspace/Task/Music/splash implementation history.
[S13]	<publication-repository>\client\phase3a_live_local_weather_v0_1\NABU_COMMAND_CENTER_LIVE_LOCAL_WEATHER_v0_1\main.c plus current 000001.map, 000001.sym and accepted LV2 binaries	Exact active NCC-SPLASH-260826-LV2 implementation source for controls, validators, scheduler, Help, splash and runtime semantics; map/sym control addresses. Source commit/hash remains pending final authorized freeze.
[S14]	SPLASH.txt + owner-approved HTML mockup	Splash visual/content authority and startup behavior.
[S15]	Owner MAME and physical-NABU regression evidence, 26 Aug 2026	Current NCC-SPLASH-260826-LV2 acceptance, photographs and final physical verification.
[S16]	NABU_Command_Center_Master_White_Paper_Codex_Review_Recommendations.txt, 26 Aug 2026	Current read-only source/Gateway/map/Git audit and correction authority for document revision 1.1.
[W01]	ADSB.lol API documentation, https://api.adsb.lol/ and https://www.adsb.lol/docs/open-data/api/ ; accessed 2026-08-26	Official point-query form /v2/point/{lat}/{lon}/{radius}, current public API/data terms; production source separately fixes 100 NM.
[W02]	USGS Earthquake real-time feeds, https://earthquake.usgs.gov/earthquakes/feed/ ; accessed 2026-08-26	Official GeoJSON Summary Feed documentation for programmatic earthquake applications.
[W03]	NOAA SWPC Data Access / JSON service, https://services.swpc.noaa.gov/json/ ; accessed 2026-08-26	Official current Space Weather JSON product service.
[W04]	Mutopia Project piece 257, https://www.mutopiaproject.org/cgi-bin/piece-info.cgi?id=257 ; accessed 2026-08-26	Production music-source record for Bach 'Aria Bist Du Bei Mir', including public-domain status and MIDI availability.
[W05]	Mido MIDI Files documentation, https://mido.readthedocs.io/en/latest/files/ ; accessed 2026-08-26	Standard MIDI-file parsing/timing semantics; production package is pinned separately to mido==1.3.3.
[W06]	NIST Internet Time Service, https://tf.nist.gov/tf-cgi/servers.cgi/en-en/ ; accessed 2026-08-26	Official generic load-balanced time.nist.gov service identity.

ID	Source	Use
[W07]	NWS API Web Service, https://www.weather.gov/documentation/services-web-api ; accessed 2026-08-26	Official api.weather.gov service, /points discovery, observation/alert API guidance and User-Agent requirement.
[W08]	U.S. Census Gazetteer ZCTA record layout, https://www.census.gov/programs-surveys/geography/technical-documentation/records-layout/gaz-record-layouts.html ; accessed 2026-08-26	Official ZCTA GEOID and representative coordinate field definitions used for ZIP-centered lookup provenance.
[W09]	WhereTheISS developer API, https://ord.wheretheiss.at/w/developer ; accessed 2026-08-26	Official developer documentation for ISS satellite id 25544 and position/velocity fields.

Table 36. Human-readable citation registry.

Appendix F - Provider / provenance checklist

Provider	v1.0 use	Release check
NIST	Time maintenance	Verify time.nist.gov service behavior and UTC semantics at release. [W06]
NWS / api.weather.gov	Weather and alerts	Verify User-Agent, schema fields, service behavior and terms/open-data status at release. [W07]
Census ZCTA + NWS points metadata	ZIP coordinate/locality resolution	Verify ZCTA coordinate dataset/service provenance and NWS locality semantics. [W08] [W07]
USGS	Earthquakes	Verify exact GeoJSON feed family/lifecycle and fields. [W02]
NOAA/SWPC	Space Weather	Verify exact JSON products and update cadence. [W03]
WhereTheISS	ISS state	Verify endpoint/terms/field behavior and ISS id 25544. [W09]
ADSB.lol	Airspace	Verify point API, current terms/licensing/attribution and production-use notice; production radius remains source-defined 100 NM. [W01]
Mutopia	Music	Verify piece 257 work metadata, public-domain status and source MIDI hash. [W04]

Table 37. Provider provenance and release checklist.

Appendix G - Release-candidate acceptance checklist

- Source BUILD_ID matches current candidate.
- 000001.bin and 000001.NABU sizes match release manifest.
- Both artifacts are byte-identical and SHA-256 matches.
- Gateway starts under the documented production Python environment.
- Required Store resources publish with expected sizes and validators accept them.

- Adapter points to correct local program folder and Store.
- MAME boots exact BUILD_ID and splash.
- Any key dismisses splash; automatic timeout works.
- Dashboard navigation and all six module maximized views work.
- ZIP edit/backspace and location refresh work.
- Automatic 60-second refresh and manual R both work independently.
- Task Manager pages show truthful LIVE/CALC/MAP/CONST/UNAVAILABLE fields.
- Music B toggle, cached load, playback and loop behavior are acceptable.
- Sound X toggle and UI cues still work.
- No regression of owner-frozen visuals.
- Physical NABU runs the same exact accepted binary.
- Physical module/navigation/reset/reload regression passes.
- Hashes, source provenance, licenses, screenshots/photos and known limitations are archived.

Appendix H - Known current limitations

Area	Current limitation / boundary
Memory	Only 646 bytes measured between BSS_END \$FC7A and initial SP \$FF00 in current candidate.
Help	Basic single-page Help exists; dense multi-page About/Help stopped at memory/key-code gate.
Diagnostics ring	Compiled out by default in splash-era source to recover memory; checkpoint/guard state remains.
Music fidelity	Two-voice bounded AY rendition; deliberately not a full multi-instrument tracker engine.
Music catalog	Single-stream v1.0 policy; no genre cycling implemented.
Physical reproducibility	Owner verified NCC-SPLASH-260826-LV2 on physical NABU, but the exact interface/driver/COM/serial-mode/cable/pinout manifest is not frozen. This paper therefore reconstructs the software and records verified physical operation; third-party interconnect replication requires Appendix L.
Final Git state	At the Codex audit, LV2 is owner-accepted but unfrozen: branch phase-page1-live-satellite-airspace-v1, HEAD dd1bad5e..., no LV2 commit/tag. Final freeze/tag requires separate Owner authorization.

Table 38. Current v1.0 limitations and explicit evidence boundaries.

Appendix I - Conclusion

NABU Command Center succeeds because it respects the machine rather than fighting it. The Z80 is not asked to be a web browser, JSON processor, MIDI workstation or geospatial server. It receives small validated records and turns them into a dense, responsive native interface. The Gateway absorbs modern volatility; the Golden transport provides a stable boundary; the client uses fixed memory and deterministic scheduling; and every milestone is tied to visible build identity, artifacts, hashes, MAME evidence and ultimately the physical NABU.

The current release candidate demonstrates the full v1.0 concept on real hardware: live terrestrial, seismic, space, orbital and airspace telemetry; dynamic locality; truthful system diagnostics; bounded background music; a native startup presentation; global refresh; and the owner-frozen graphical Command Center shell. The remaining release work is primarily preservation: final freeze/tag, manifest, licensing/provenance audit, public packaging and clean-machine reproduction.

The engineering standard remains intentionally conservative: build upon verified layers, change one principal variable at a time, never invent the machine, and let evidence - not plausible-looking code - decide when a NABU program is finished.

Appendix J - Codex Technical Audit Disposition

The 26 Aug 2026 Codex review classified the v1.0 draft as MAJOR UPDATE REQUIRED. This document revision applies the nine MUST corrections, incorporates the nine SHOULD updates where the current evidence is available, and adds the reconstruction controls below. Items that require Owner-only physical hardware knowledge or an authorized final Git freeze remain explicitly unresolved rather than guessed. [S16]

Review item	Disposition	Document action
WP-MUST-001	Applied	Authority hierarchy replaced with current review hierarchy.
WP-MUST-002	Applied	Exact production zcc command includes minimal_vdp_isr.asm, -m, -s and --list; stale BUILD.BAT echo caveat recorded.
WP-MUST-003	Applied	ncc_wxhist.dat corrected to 64 B, Gateway -> NABU.
WP-MUST-004	Applied	ZIP, TIME, WX, WXH1 and WA01 contracts added; Appendix D expanded.
WP-MUST-005	Applied	[S13] changed to exact active LV2 path/source-map-artifact evidence; freeze hash/commit left pending.
WP-MUST-006	Applied	Milestone ledger rebuilt without treating abbreviated digests as release identity.
WP-MUST-007	Applied with explicit boundary	Physical-interconnect manifest added; missing hardware values remain UNVERIFIED.
WP-MUST-008	Applied	Exact MAME commands/path, Adapter Store/program-load roles and startup order documented.
WP-MUST-009	Applied	LV2 consistently labeled owner-accepted, unfrozen working-tree release candidate.
WP-SHOULD-001..009	Applied / bounded	Gateway/provider map, preflight, diagnostics scope, raw key codes, refresh queue semantics, Music C-key clarification, Task Manager provenance, figure-evidence caption and citation access dates updated. Exact refresh call order remains a final-freeze source-transcription item; it is not inferred.

Appendix K - Release Manifest Template

Populate this manifest only from the final Owner-authorized freeze. It is intentionally machine-readable in concept but shown here as a human table so missing evidence is visible.

Manifest key	Release value / required evidence
product	NABU Command Center v1.0
build_id	NCC-SPLASH-260826-LV2
artifact_bin	000001.bin
artifact_nabu	000001.NABU
artifact_size	59,501 bytes each
sha256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
source_commit	UNVERIFIED / pending final freeze
release_tag	NONE / pending final freeze
branch	phase-page1-live-satellite-airspace-v1 at Codex audit
z88dk	2.4 baseline; installed exact version/hash to capture at freeze
mame	0.250; exact executable path in Section 39
internet_adapter	NABU-Internet-Adapter-84.exe; MAME-era verified version 2025.02.07.00, recheck at freeze
gateway_python	UNVERIFIED until release manifest captures actual executable/version
mido	1.3.3 pinned by requirements-music.txt
mame_status	OWNER VERIFIED
physical_status	OWNER VERIFIED
physical_manifest	REQUIRED for third-party interconnect reconstruction

Appendix L - Physical Deployment Manifest Template

The following fields must be completed from the Owner's actual hardware before the paper can claim third-party physical interconnect reproducibility. No field may be populated from model memory.

Physical field	Verified value
NABU model / board revision	UNVERIFIED
NABU serial/HCCA connector revision	UNVERIFIED
USB-to-RS-422 adapter manufacturer/model	UNVERIFIED
Interface chipset	UNVERIFIED
Windows driver/version	UNVERIFIED
Observed COM port	UNVERIFIED
Adapter software version	UNVERIFIED for physical-mode reproduction
Adapter physical serial mode/settings	UNVERIFIED
Cable connector types	UNVERIFIED
Cable pinout / continuity record	UNVERIFIED
Electrical mode / termination details	UNVERIFIED
Power/connect/disconnect procedure	UNVERIFIED

Physical field	Verified value
Accepted BUILD_ID	NCC-SPLASH-260826-LV2
Accepted SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
Physical acceptance date	2026-08-26 owner report